Building Interoperable Location-based Augmented Reality with the Open AR Cloud
ISMAR 2026 Tutorial - October 5, 2026, 08:15–12:00, Sezione 6+7

Reality Modeling Project

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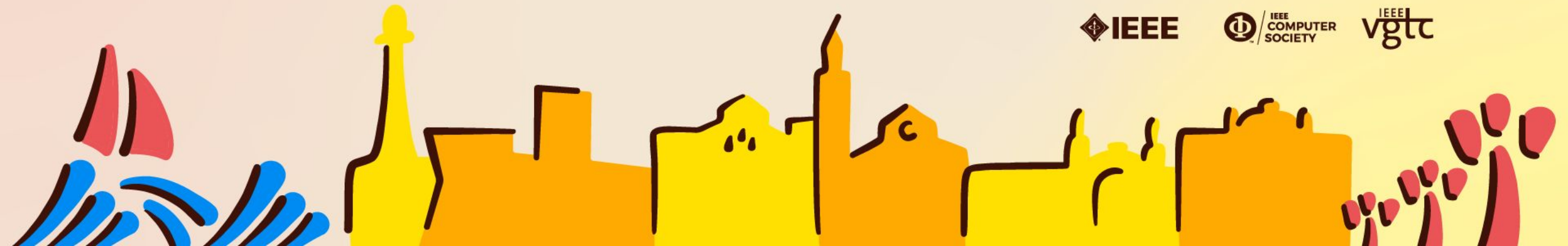
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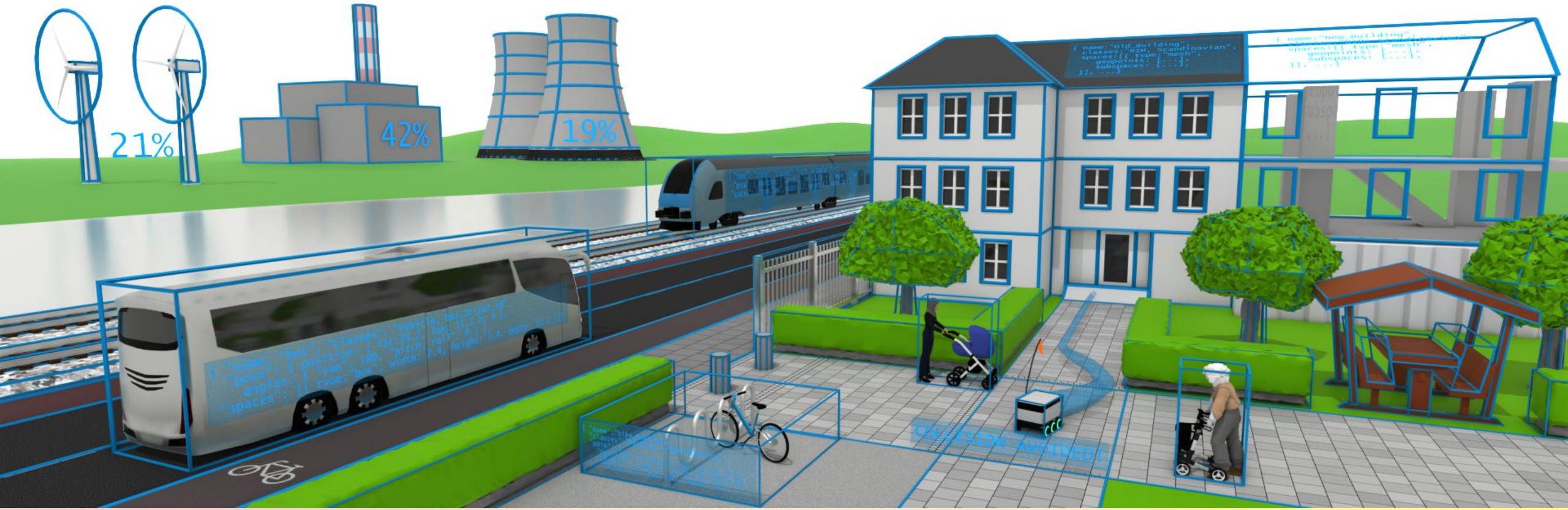
The 25th International Symposium on Mixed and Augmented Reality



Introduction

What's this about?

The Reality Modeling Project aims to provide a common framework to describe the physical environment in an open, federated and interoperable way, enabling the creation of advanced spatial-based interaction techniques for XR.



Motivation

Why does this matter?

Currently, there are three main issues hindering the creation of a common language to describe reality:

- **Lack of Interoperability:** While there have been many attempts to define a common way to describe reality (e.g., buildingSMART BIM, OGC CityGML, etc.), the associated conceptual models are restricted to specific knowledge fields, resulting in incompatible file formats, loss of data and… a lot of headaches.
- **Limited Spatial Information:** Most 3D engines and editors operate with the concept of “scenes”, where the logical relationships between entities are restricted to simple tree structures (with transformation chains), restricting spatial information to bounding boxes, colliders or navigation meshes.
- **Poor User Interaction:** Since there is not a reliable representation of reality, XR applications are often constrained to pre-defined boundaries for safety reasons, while robotic systems have to rely on sensors to constantly evaluate their surroundings (a “scene understanding”).

Features

The three main features of the Reality Modeling framework are:

- **Open World Model:** Transcending domain-specific limitations, this conceptual model allows the collaborative description of physical entities, combining multiple visual and logic representations to achieve a more comprehensive description of the real world.
- **(Geo)Spatial Understanding:** Apart from its pose (location and orientation), each entity in the Open World Model also contains information about their associated spaces (including those generated by gravity or the relative position of the user) and how they are interconnected together.
- **Enhanced User Interaction:** The additional semantic and spatial information removes the need to create safety boundaries for XR experiences, while also allowing the definition of more intuitive and ergonomic user interaction techniques that adapt to the physical environment and the body of the users.

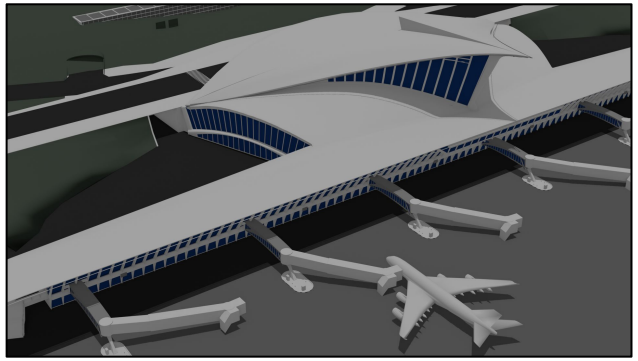
Architecture

The Reality Modeling framework provides three software layers (intended to be integrated on top of existing 3D middleware solutions):

- **Semantic Description Layer:** Enables the dynamic definition of new semantic domains, classes and properties, as well as advanced relationships between them (thus, enabling the creation of mechanisms to translate between existing conceptual models and serialization systems).
- **Spatial Computing Layer:** Aside of allowing the definition of spaces with complex 3D geometries, this layer provides a collection of algorithms to compute advanced spatial operations for projection/CRS transformations, rule-based navigation techniques, and real-time simulations.
- **User Interaction Layer:** Beyond providing mechanisms to visually represent the semantic and spatial data of the entities in an easily understandable way, this layer provides a collection of user interaction techniques (and widgets) to easily create 3D user interfaces.

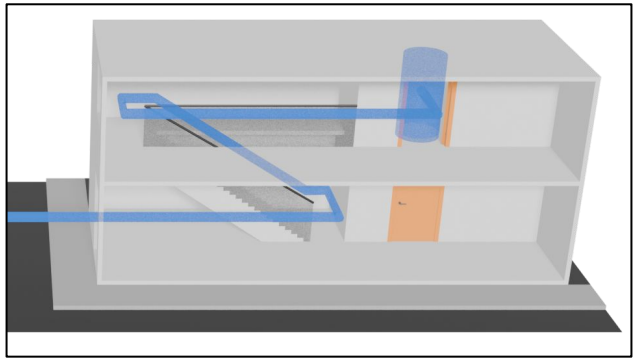
Examples

Among the numerous applications, the following three use cases provide a comprehensive view of the possibilities of this technology:



Built Environment

The geospatial layer makes it possible to create large and semantically-rich Digital Twins, seamlessly connecting entities across the planet (including satellites orbiting it.



Robotic Systems

Better pathfinding systems allow autonomous vehicles and drones to navigate the real world and collaborate together while avoiding dangerous collisions with other entities (and users).



Augmented Reality

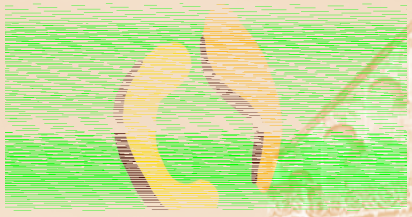
A good definition of the physical environment does not only allow the creation of beautiful 3D maps, but also enables the integration of virtual objects into it from a first person perspective.

Contribute

We need your help!

If you want to contribute to this project, you can do so in different ways:

- **(Re)Implementations:** Help us translate these ideas into code and make the it available for existing 3D engines (e.g., ThreeJS, Cesium, Unreal, etc).
- **Standardization:** Participate in OGC meetings to define new geospatial standards that enables a better description of the real world entities.
- **Applications:** To properly validate the framework, it is necessary to test it in real-world scenarios, so... if there is an interesting project, please let us know.



Questions? Collaboration?



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